

Part IX — Modern Physics and the Universe · Chapter 49

Quantum Ideas

The physics of the very small: probability, uncertainty, and quantised states

Classical physics describes a world of certainty and continuity. A ball thrown through the air follows a perfectly predictable parabola. An electron fired at a detector will hit a specific point. Light waves spread smoothly through space and carry energy that can take any value. Or so it seemed. In the early twentieth century, a series of experiments shattered these assumptions. Light came in indivisible packets. Electrons diffracted like waves. Particles could not have precisely defined positions and momenta simultaneously. At the smallest scales, nature is not a clockwork machine but a probabilistic dance of allowed states and quantum jumps. This chapter explores those ideas — not with heavy mathematics, but with the conceptual clarity needed to understand what quantum physics actually says about the world, why it is so different from classical physics, and how it underlies every electronic device, every MRI machine, and every laser in modern life.

49.1 What “Quantum” Means

The word quantum comes from the Latin for “how much” or “a discrete amount.” In physics, a quantum is the smallest possible discrete unit of a physical quantity. Quantum physics is the physics of systems where physical quantities are quantised — restricted to specific discrete values rather than varying continuously.

In everyday life, most things appear continuous. Temperature, velocity, position, electrical current — all seem to vary smoothly and can take any value within a range. This is a correct description at large scales. But at atomic and subatomic scales, many quantities are found to be discrete. Energy, angular momentum, electric charge — all come in specific minimum units. You cannot have half a unit of charge. You cannot have 2.5 units of angular momentum. You cannot have an electron in a hydrogen atom with energy of -6.0 eV.

Quantum physics was developed between roughly 1900 and 1930 through contributions from Planck, Einstein, Bohr, de Broglie, Heisenberg, Schrödinger, Dirac, and many others. Its predictions have been confirmed to extraordinary precision — the anomalous magnetic moment of the electron is predicted and measured to within one part in 10^4 . No prediction of quantum mechanics has ever been proven wrong by experiment. It is, by the measure of experimental verification, the most successful physical theory ever devised.

Key Point: Quantum physics does not replace classical physics — it contains classical physics as a limiting case. For objects much larger than atoms, quantum effects average out and classical physics is an excellent approximation. The transition from quantum to classical behaviour is called the correspondence principle: in the limit of large quantum numbers, quantum predictions approach classical predictions.

49.2 Quantised Energy

The first clear indication that energy is quantised came from Max Planck's work on blackbody radiation in 1900. A blackbody is an idealised object that absorbs all electromagnetic radiation falling on it and emits radiation based only on its temperature. The spectrum of blackbody radiation (the distribution of emitted power across wavelengths) was well-measured but could not be explained by classical physics, which predicted that a hot object should emit infinite energy at short wavelengths — an absurd result called the "ultraviolet catastrophe."

Planck found that he could explain the observed spectrum only by assuming that the energy of each oscillator in the blackbody is quantised: $E = nhf$, where n is a non-negative integer, f is the oscillator's frequency, and h is a new fundamental constant now called Planck's constant.

$$E = nhf \quad (n = 0, 1, 2, 3, \dots)$$

Planck's quantisation condition. $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ (Planck's constant); f = frequency (Hz); n = non-negative integer. Energy can only be $0, hf, 2hf, 3hf, \dots$ never a fractional multiple of hf .

Planck himself was uncomfortable with this idea and tried to find a classical interpretation. He could not. The quantisation was real, and it led directly to all of quantum physics. Planck's constant h is the fundamental scale of quantum effects: when h can be neglected (i.e. when energies are much larger than hf), classical physics applies. At atomic scales, hf is comparable to typical energies, and quantum effects dominate.

Historical Note: Max Planck (1858–1947) described his introduction of energy quantisation as "an act of desperation." He had tried every classical approach and none worked. He was a deeply conservative physicist and expected someone to find a classical explanation. No one ever did. Planck received the Nobel Prize in Physics in 1918 for this discovery.

49.3 Photons and Light Quanta

In 1905, Albert Einstein took Planck's quantisation further. Planck had quantised the energies of the oscillators emitting and absorbing radiation, but still treated light itself as a continuous wave. Einstein proposed something more radical: light itself is composed of discrete packets of energy called photons (a term introduced later by Gilbert Lewis in 1926).

Each photon carries energy proportional to the frequency of the light:

$$E = hf = hc / \lambda$$

Photon energy. $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$; f = frequency (Hz); $c = 3.00 \times 10^8 \text{ m/s}$; λ = wavelength (m). Higher-frequency light (blue, UV, X-ray) has higher-energy photons than lower-frequency light (red, infrared, microwave).

Photons have no rest mass. They always travel at c in vacuum. Each photon carries both energy $E = hf$ and momentum $p = h/\lambda = hf/c$. Despite having zero rest mass, photons have momentum — a fact confirmed by the radiation pressure exerted by light on surfaces.

Example — Photon energy at different wavelengths Red light ($\lambda = 700 \text{ nm}$): $E = hc/\lambda = (6.626 \times 10^{-34} \times 3.00 \times 10^8)/(700 \times 10^{-9}) = 2.84 \times 10^{-19} \text{ J} = 1.77 \text{ eV}$. Blue light ($\lambda = 400 \text{ nm}$): $E = (6.626 \times 10^{-34} \times 3.00 \times 10^8)/(400 \times 10^{-9}) = 4.97 \times 10^{-19} \text{ J} = 3.10 \text{ eV}$. A blue photon carries 1.75× more energy than a red photon.

Table 49.1 Photon energies across the electromagnetic spectrum.

49.4 The Photoelectric Effect Revisited

The photoelectric effect (introduced in earlier chapters) is the emission of electrons from a metal surface when light shines on it. Classical physics expected: (1) any frequency of light should work, given enough intensity; (2) brighter light should produce more energetic electrons; (3) energy should build up gradually, so there should be a time delay before emission begins.

Experiments showed exactly the opposite: (1) only light above a threshold frequency causes emission, regardless of intensity; (2) brighter light produces more electrons but not more energetic ones — the electron energies depend only on frequency; (3) emission begins instantaneously, with no time delay even at very low intensities.

Einstein's photon model explains all three observations. Each electron is ejected by a single photon. The photon must have enough energy to overcome the work function ϕ (the binding energy of the electron to the metal). If $hf < \phi$, no electron is ejected regardless of intensity (there are many low-energy photons but each individual one cannot liberate an electron). If $hf > \phi$, the excess energy becomes the electron's kinetic energy:

$$KE_{\max} = hf - \phi$$

Photoelectric equation (Einstein, 1905). $KE_{\max}$ = maximum kinetic energy of ejected electron (J); hf = photon energy (J); ϕ = work function of the metal (J or eV). The threshold frequency is $f_0 = \phi/h$.

Einstein received the 1921 Nobel Prize in Physics specifically for this explanation of the photoelectric effect, not for relativity. It was the decisive experimental confirmation of the photon concept. Photoelectric cells, solar panels, and photomultiplier tubes all rely on this principle.

Example — Photoelectric effect Sodium has work function $\phi = 2.28$ eV. Light of wavelength 400 nm shines on it. Photon energy: $E = hc/\lambda = 1240 \text{ eV}\cdot\text{nm} / 400 \text{ nm} = 3.10$ eV. (Note: $hc = 1240 \text{ eV}\cdot\text{nm}$ is a useful constant to remember.) $KE_{\max} = 3.10 - 2.28 = 0.82$ eV. Threshold wavelength: $\lambda_0 = hc/\phi = 1240/2.28 = 544$ nm (green light). Light longer than 544 nm cannot eject electrons from sodium.

49.5 Matter Waves

If light — classically a wave — can behave like a particle (a photon), could particles — classically objects with definite positions and momenta — behave like waves? In 1924, Louis de Broglie proposed exactly this. He suggested that every particle with momentum p has an associated wavelength:

$$\lambda = h / p = h / (mv)$$

de Broglie wavelength. λ = wavelength (m); $h = 6.626 \times 10^{-34}$ J·s; p = momentum (kg·m/s); m = mass (kg); v = speed (m/s). Every particle has a wave nature characterised by this wavelength.

For macroscopic objects, the de Broglie wavelength is negligibly small. A 1 kg ball moving at 1 m/s has $\lambda = h/mv = 6.626 \times 10^{-34}$ m — far smaller than any atom, completely undetectable. But for an electron moving at typical atomic speeds ($\sim 10^6$ m/s), $\lambda \approx 10^{-10}$ m $\approx$ 0.1 nm, which is comparable to the size of an atom and the spacing of atoms in a crystal. This should produce detectable wave effects.

In 1927, Davisson and Germer fired electrons at a nickel crystal and observed diffraction patterns — exactly as expected for waves with the de Broglie wavelength. This was direct experimental proof that electrons have wave properties. Shortly after, G. P. Thomson (son of J. J. Thomson, who established the particle nature of electrons) independently observed electron diffraction through thin metal films. Father proved electrons are particles; son proved they are also waves.

Example — de Broglie wavelength of an electron An electron is accelerated through 54 V. Its kinetic energy is $KE = 54 \text{ eV} = 54 \times 1.602 \times 10^{-19} = 8.65 \times 10^{-18}$ J. Momentum: $p = \sqrt{2m_e KE} = \sqrt{2 \times 9.109 \times 10^{-31} \times 8.65 \times 10^{-18}} = 3.97 \times 10^{-24}$ kg·m/s. $\lambda = h/p = 6.626 \times 10^{-34} / 3.97 \times 10^{-24} = 0.167$ nm. This is the Davisson–Germer experimental wavelength — the calculation confirmed their measurement.

49.6 Wave-Particle Duality

Quantum objects — electrons, photons, atoms, even molecules — exhibit both wave and particle behaviour. This is wave-particle duality. It is not that quantum objects are sometimes waves and sometimes particles; it is that they are neither, in the classical sense, but something new that has aspects of both.

The double-slit experiment illustrates this beautifully. Fire electrons one at a time at a barrier with two slits. Each electron arrives as a localised dot on the detector — particle-like. But over time, the dots accumulate into an interference pattern of alternating bright and dark bands — wave-like. The interference pattern appears even when electrons pass through one at a time, meaning each electron's probability of landing at a point is determined by the wave nature of the electron — it “interferes with itself” by passing through both slits simultaneously.

If you try to determine which slit the electron went through by placing a detector at one slit, the interference pattern disappears. The act of measurement disturbs the system. This is not a failure of instrumentation — it is a fundamental feature of quantum mechanics.

The Rule: Wave-particle duality: quantum objects do not have a definite nature (wave or particle) independent of measurement. The experimental setup determines which aspect is revealed. Both aspects are real; neither alone is the complete picture.

This duality is not limited to electrons and photons. Neutrons, protons, atoms, and even molecules as large as C_{60} (buckminsterfullerene, containing 60 carbon atoms) have been shown to produce interference patterns in double-slit experiments. Wave behaviour is universal in quantum systems.

49.7 Electron Probability Clouds

The modern quantum mechanical description of an electron in an atom replaces Bohr's fixed circular orbits with probability distributions. The electron does not travel on a definite path; instead, its state is described by a wave function ψ (psi), and the probability of finding the electron in a small volume dV at position r is proportional to $|\psi|^2 dV$.

The probability distribution $|\psi|^2$ describes an electron cloud: a three-dimensional region where the electron is likely to be found. For the ground state of hydrogen ($n = 1$), the probability is highest at the Bohr radius (0.053 nm) from the nucleus and decreases exponentially with distance. The electron is most likely to be near the nucleus but can be found at any distance with non-zero probability — including, in principle, on the other side of the universe, though the probability becomes vanishingly small.

Different energy levels correspond to different orbital shapes. The s-orbitals ($l = 0$) are spherically symmetric. The p-orbitals ($l = 1$) are dumbbell-shaped with two lobes. The d-orbitals ($l = 2$) have four lobes. These orbital shapes determine how atoms bond to form molecules and are the basis of all chemistry.

Key Point: The electron cloud is not a picture of an electron smeared out in space — when you measure the position of an electron, you always find it at a single, specific location. The cloud shows where that location is likely to be. Before measurement, the electron does not have a definite position; it has a probability distribution. After measurement, it has a definite (if temporary) position.

49.8 The Uncertainty Principle

In 1927, Werner Heisenberg derived one of the most profound results in all of physics: the Heisenberg Uncertainty Principle. It states that certain pairs of physical quantities — called conjugate variables — cannot both be known precisely at the same time. The more precisely one is known, the less precisely the other can be known.

The most important conjugate pair is position and momentum:

$$\Delta x \cdot \Delta p_x \geq \hbar/2$$

Heisenberg Uncertainty Principle for position and momentum. Δx = uncertainty in position (m); Δp_x = uncertainty in momentum in the x-direction (kg·m/s); $\hbar = h/(2\pi) = 1.055 \times 10^{-34}$ J·s (reduced Planck constant). The product of the two uncertainties can never be less than $\hbar/2$.

There is also an energy-time uncertainty relation: $\Delta E \cdot \Delta t \geq \hbar/2$. This has an important consequence: a quantum state that exists for a short time Δt has an intrinsic energy uncertainty $\Delta E \geq \hbar/(2\Delta t)$. This is why spectral lines have finite widths rather than being perfectly sharp — excited states have finite lifetimes Δt , giving an intrinsic energy spread that broadens the emitted spectral line.

The uncertainty principle is not due to clumsy measurement. It is not a statement about the limitations of instruments. It is a fundamental feature of nature: conjugate variables do not simultaneously have definite values. An electron does not have both a definite position and a definite momentum. Asking for both is like asking for the north edge of a cloud.

Example — Uncertainty principle estimate An electron is confined to a region of size $\Delta x = 0.1$ nm (roughly atomic scale). Minimum momentum uncertainty: $\Delta p \geq \hbar/(2\Delta x) = (1.055 \times 10^{-34})/(2 \times 10^{-10}) = 5.3 \times 10^{-25}$ kg·m/s. Minimum kinetic energy: $KE \geq (\Delta p)^2/(2m_e) = (5.3 \times 10^{-25})^2 / (2 \times 9.1 \times 10^{-31}) \approx 1.5 \times 10^{-19}$ J ≈ 1 eV. Electrons confined to atoms must have kinetic energies of order 1 eV, consistent with atomic energy scales.

49.9 Energy Levels and Quantum States

In the full quantum mechanical treatment, an electron in an atom is described by four quantum numbers: the principal quantum number n (energy level), the angular momentum quantum number l (orbital shape), the magnetic quantum number m_l (orbital orientation), and the spin quantum number m_s (electron spin, either $+1/2$ or $-1/2$).

Each unique set of quantum numbers specifies a quantum state. The Pauli exclusion principle (Wolfgang Pauli, 1925) states that no two electrons in the same atom can have the same set of four quantum numbers. This means each quantum state can hold at most one electron.

The filling of quantum states according to the Pauli exclusion principle, with states filled from lowest energy upward, explains the structure of the periodic table. Hydrogen has one electron; helium fills the $n = 1$ shell; lithium begins the $n = 2$ shell; neon fills the $n = 2$ shell; and so on. The chemical properties of elements — valence, reactivity, bonding — are determined by the arrangement of electrons in the outermost partially filled shell.

n	l range	Orbital names	Max electrons shell	Elements in period
1	0	1s	2	H, He
2	0, 1	2s, 2p	8	Li to Ne
3	0, 1, 2	3s, 3p, 3d	18	Na to Ar (simplified)
4	0, 1, 2, 3	4s, 4p, 4d, 4f	32	K to Kr (simplified)

Table 49.2 *Quantum numbers and electron capacity for each principal shell. Each orbital holds 2 electrons (spin up and spin down).*

49.10 Quantum Transitions

A quantum transition is an abrupt change of a system from one quantum state to another. Unlike classical transitions, which can be gradual, quantum transitions are instantaneous: the system is in one state, then suddenly in another. There is no in-between state.

Transitions between energy levels in atoms occur by the absorption or emission of photons (or by other processes such as electron collisions). The selection rules of quantum mechanics specify which transitions are allowed and which are forbidden. Not every transition between any two levels is permitted; the angular momentum quantum number l must change by exactly ± 1 in an electric dipole transition (the most common type). Forbidden transitions are not truly impossible but are much less likely.

Stimulated emission is a type of quantum transition with enormous practical importance. In spontaneous emission, an excited atom emits a photon at a random time and in a random direction. In stimulated emission, a passing photon with energy equal to the transition energy triggers the atom to emit an identical photon — same energy, same direction, same phase. This is the mechanism behind the laser (Light Amplification by Stimulated Emission of Radiation).

Quantum tunnelling is another type of quantum behaviour with no classical analogue. A particle encountering a potential energy barrier higher than its kinetic energy would be completely reflected classically. In quantum mechanics, the wave function extends into and through the barrier, giving a non-zero probability that the particle will appear on the other side. Tunnelling underlies alpha decay, the operation of tunnel diodes, the scanning tunnelling microscope, and the nuclear fusion reactions in the Sun.

49.11 Quantum Models vs. Classical Models

Classical physics and quantum physics make different predictions at different scales. Understanding when to use each is important.

Aspect	Classical physics	Quantum physics
Particle position	Definite, predictable trajectory	Probability distribution; uncertain until measured
Energy	Continuous; any value allowed	Quantised; only specific values allowed
Light	Continuous electromagnetic wave	Stream of discrete photons with $E = hf$
Measurement	Observation does not disturb system	Measurement changes the quantum state
Electron in atom	Spirals inward; no stable orbits	Occupies stable quantised energy levels
Barrier crossing	Forbidden if energy < barrier height	Quantum tunnelling gives non-zero probability
Predictability	Deterministic; given initial conditions, future is certain	Probabilistic; only probabilities can be predicted
Scale of validity	Works for macroscopic objects	Required for atoms, molecules, subatomic particles

Table 49.3 *Key differences between classical and quantum descriptions of physical systems.*

Classical physics emerges from quantum physics in the limit of large quantum numbers (the correspondence principle) and when decoherence — the interaction of quantum systems with their environment — destroys quantum superpositions and produces effectively classical behaviour. Everyday objects contain $\sim 10^{23}$ atoms, and quantum fluctuations average out to negligible levels. For individual atoms, molecules, electrons, and photons, quantum effects are essential.

49.12 Quantum Physics in Modern Technology

Quantum physics is not merely abstract theory. It underlies virtually all modern technology.

Semiconductors and transistors

The operation of transistors — the building blocks of all computers, smartphones, and digital devices — is entirely quantum mechanical. The band structure of semiconductors (allowed and forbidden energy bands for electrons in a crystal lattice) arises from quantum mechanics. The behaviour of n-type and p-type doping, the p-n junction, and the field-effect transistor all rely on quantum energy levels and probability distributions.

Lasers

Lasers produce coherent, monochromatic, directional light through stimulated emission of photons (Section 49.10). A laser medium (gas, liquid, solid, or semiconductor) is pumped to create a population inversion (more atoms in an excited state than in the lower state), allowing stimulated emission to dominate. Lasers are used in surgery, communications, manufacturing, CD/DVD/Blu-ray players, barcode scanners, range-finding, and scientific instruments.

MRI and NMR

Magnetic resonance imaging (MRI) exploits the quantum spin of protons (hydrogen nuclei) in the body. Proton spin has two quantised states in a magnetic field; radio-frequency photons with energy matching the energy difference between states cause transitions whose characteristics reveal tissue structure. MRI provides detailed images of soft tissue without ionising radiation.

LED and solar cells

Light-emitting diodes (LEDs) emit photons when electrons cross the band gap of a semiconductor: each photon energy equals the band gap energy, giving a specific colour. Solar cells work in reverse: photons with energy above the band gap excite electrons, creating electrical current. Both devices are direct applications of photon-electron energy level interactions.

Atomic clocks

Atomic clocks use the precisely defined transition frequency between two hyperfine energy levels of caesium-133 atoms (9,192,631,770 Hz, by definition) as a frequency standard. The GPS system depends on atomic clocks; without quantum-mechanical precision, GPS position errors would accumulate at kilometres per day.

49.13 Common Misunderstandings About Quantum Physics

- The uncertainty principle is not about measurement disturbance. The principle is often explained as “measuring position disturbs momentum.” While some disturbance does occur, the principle is deeper: position and momentum are not simultaneously defined properties of a quantum system, independent of measurement. The uncertainty is fundamental, not instrumental.
- Wave-particle duality does not mean objects switch between wave and particle. It means quantum objects have properties of both simultaneously. The apparent contradiction disappears when you recognise that “wave” and “particle” are classical concepts that do not perfectly map onto quantum reality.
- Quantum randomness is not the same as ignorance. Classical randomness (a coin flip) arises because we do not know the initial conditions precisely enough. Quantum randomness (the exact moment an excited atom decays) is fundamental: there is no hidden information that, if known, would allow prediction. This was established by Bell’s theorem and confirmed by experiment.
- “Observer” in quantum mechanics does not mean a conscious being. Any physical interaction that correlates the quantum system with the measuring apparatus constitutes a “measurement.” Consciousness is irrelevant. A rock interacting with an electron is an “observer” in the quantum sense.
- Schrödinger’s cat is a thought experiment illustrating the absurdity of applying quantum superposition to macroscopic objects, not a serious proposal about cat physics. Real cats are macroscopic, classical objects; decoherence prevents the superposition from surviving to macroscopic scales. Schrödinger intended his thought experiment as a *reductio ad absurdum* critique of naive quantum measurement theory.
- Quantum mechanics does not say “anything is possible.” Quantum tunnelling has a calculable, often extremely small probability. The uncertainty principle gives ranges, not permission for arbitrary behaviour. Quantum physics is precisely quantitative, not a licence for mysticism.

Common Mistake: Photon energy problems commonly go wrong when students confuse wavelength units. $E = hc/\lambda$ requires λ in metres. If λ is given in nm, multiply by 10^{-9} . A useful shortcut: $E \text{ (eV)} = 1240 / \lambda \text{ (nm)}$. Memorising this avoids unit conversion errors in photon energy calculations.

Chapter Summary

Quantum physics describes the behaviour of matter and energy at atomic and subatomic scales, where physical quantities are quantised, particles exhibit wave behaviour, and probabilities replace certainties.

- Quantised energy: energy comes in discrete packets (quanta). Planck's constant $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ sets the scale of quantum effects.
 - Photons: light consists of discrete photons with energy $E = hf = hc/\lambda$. The photoelectric effect confirms that energy is transferred in single photon-electron interactions.
 - de Broglie waves: every particle has a wavelength $\lambda = h/p$. Electron diffraction experiments confirm matter wave behaviour.
 - Wave-particle duality: quantum objects exhibit both wave and particle properties depending on the measurement. The double-slit interference pattern persists even for single particles.
 - Probability clouds: the electron in an atom is described by a wave function; $|\psi|^2$ gives the probability distribution. Orbitals are probability clouds, not fixed paths.
 - Uncertainty principle: $\Delta x \cdot \Delta p \geq \hbar/2$. Position and momentum cannot both be precisely defined. This is fundamental, not a measurement limitation.
 - Quantum technology: semiconductors, lasers, MRI, LEDs, solar cells, and atomic clocks all depend directly on quantum mechanical principles.
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Key Terms

quantum: The smallest discrete unit of a physical quantity; e.g. a photon is a quantum of electromagnetic energy.

Planck's constant (h): $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$; the fundamental constant of quantum physics; sets the scale of quantisation.

photon: A quantum of electromagnetic radiation; energy $E = hf = hc/\lambda$; zero rest mass; always travels at c .

work function (ϕ): The minimum energy required to remove an electron from a metal surface; determines the threshold frequency for the photoelectric effect.

photoelectric effect: Emission of electrons from a metal surface when light above a threshold frequency strikes it; explained by Einstein using photons.

de Broglie wavelength: $\lambda = h/p$; the wavelength associated with a particle of momentum p ; confirmed by electron diffraction.

wave-particle duality: The property of quantum objects (electrons, photons, etc.) of exhibiting both wave and particle behaviour, depending on the experimental context.

wave function (ψ): A mathematical function describing the quantum state of a particle; $|\psi|^2$ gives the probability distribution for finding the particle at a given location.

Heisenberg Uncertainty Principle: $\Delta x \cdot \Delta p \geq \hbar/2$; the product of uncertainties in position and momentum has a fundamental minimum; not due to measurement disturbance.

principal quantum number (n): Quantum number $n = 1, 2, 3, \dots$ specifying the energy level of an electron in an atom; determines the shell.

Pauli exclusion principle: No two electrons in the same atom may have the same set of four quantum numbers; limits each quantum state to one electron.

stimulated emission: The process by which an incoming photon triggers an excited atom to emit an identical photon (same energy, phase, and direction); the basis of laser operation.

quantum tunnelling: The quantum mechanical phenomenon in which a particle passes through a potential energy barrier higher than its kinetic energy; has no classical analogue.

blackbody radiation: The electromagnetic radiation emitted by an ideal absorber at thermal equilibrium; explained by Planck's quantisation hypothesis (1900).

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. Explain the ultraviolet catastrophe. What did classical physics predict about blackbody radiation at short wavelengths, and why was this result impossible? How did Planck resolve the problem?
 2. Describe the photoelectric effect experiment. State three observations that cannot be explained by classical wave theory and explain how the photon model resolves each one.
 3. Explain de Broglie's hypothesis. Why are wave properties of matter not observed for everyday objects such as a cricket ball, but are easily observed for electrons?
 4. Describe the double-slit experiment with electrons. What result is obtained, and what does it reveal about wave-particle duality? What happens when you try to detect which slit the electron passes through?
 5. Explain the Heisenberg Uncertainty Principle in your own words. Give an example of a calculation showing why atomic confinement implies high kinetic energies.
 6. Explain what the wave function represents and what $|\psi|^2$ means physically. In what sense is an electron's position "undefined" before measurement?
 7. Explain the Pauli exclusion principle and explain how it is responsible for the structure of the periodic table.
 8. Choose one modern technology (laser, semiconductor, MRI, LED, or atomic clock) and explain in detail how it depends on quantum mechanical principles.
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Photon Energy Problems

9. Find the energy in joules and electron-volts of a photon of (a) green light (520 nm), (b) X-ray radiation (0.10 nm), and (c) FM radio waves (3.0 m wavelength).
10. A photon has energy 4.5 eV. Find its (a) frequency in Hz, (b) wavelength in nm, and (c) region of the electromagnetic spectrum (use Table 49.1).
11. A laser pointer emits red light at 650 nm with power 5.0 mW. (a) Find the energy per photon. (b) Find the number of photons emitted per second. (c) If 1% of these photons are absorbed by the retina, how many photons enter the eye per second?
12. Ultraviolet radiation of wavelength 200 nm strikes a platinum surface (work function $\phi = 5.65$ eV). (a) Find the photon energy. (b) Does the photoelectric effect occur? (c) If so, find the maximum kinetic energy of ejected electrons and the stopping voltage needed to halt them.

Energy Level Exercises

13. For hydrogen: use $E_n = -13.6/n^2$ eV. (a) Find the energy of the photon emitted when an electron drops from $n = 4$ to $n = 1$. (b) Is this photon visible, UV, or IR? (c) What quantum mechanical series does this belong to?
14. A hydrogen atom absorbs a photon and its electron jumps from $n = 1$ to $n = 3$. (a) Find the energy of the absorbed photon in eV. (b) Find the wavelength of the photon. (c) From $n = 3$, list all possible single-step downward transitions and calculate the wavelength of each emitted photon.
15. An electron has de Broglie wavelength 0.25 nm. (a) Find its momentum. (b) Find its kinetic energy in eV. (c) Through what potential difference V must it be accelerated (from rest) to achieve this wavelength?
16. An excited state in a hydrogen atom has a lifetime of $\Delta t = 1.0 \times 10^{-8}$ s. Use $\Delta E \cdot \Delta t \geq \hbar/2$ to estimate the minimum energy uncertainty of this level in eV. What width (in nm) does this give to a spectral line at 656 nm?

Quantum Concept Questions

17. Explain why quantum tunnelling is impossible classically but possible quantum mechanically. Give one example of a technology or natural phenomenon that depends on quantum tunnelling.
18. A student says: "Electrons are waves, not particles, because they produce interference patterns." Explain what is correct and what is oversimplified about this statement.
19. The Pauli exclusion principle prevents two electrons from occupying the same quantum state. How does this explain why the $n = 1$ shell holds exactly 2 electrons and the $n = 2$ shell holds exactly 8? Show the quantum number combinations explicitly.
20. Explain why the energy-time uncertainty relation implies that spectral lines have finite widths rather than being perfectly sharp. Would a longer-lived excited state give a narrower or wider spectral line? Justify your answer.

Modern Technology Applications

21. An LED emits orange light at 615 nm. (a) Find the photon energy in eV. (b) This photon energy equals the band gap of the semiconductor. If an electron falls from the conduction band to the valence band across this gap, what is the band gap energy? (c) Why can't this LED emit green light (520 nm) without changing the semiconductor material?
22. A laser is made from a semiconductor with band gap 1.43 eV (gallium arsenide). (a) Find the wavelength of light emitted. (b) Is this visible? (c) The laser outputs 1.0 mW. How many photons per second does it emit?
23. MRI uses the energy splitting of proton spin states in a 3.0 T magnetic field. The energy difference is $\Delta E = 2\mu_p B$, where $\mu_p = 1.41 \times 10^{-26}$ J/T. (a) Find ΔE in joules. (b) Find the radio frequency $f = \Delta E/h$ needed to drive transitions. (c) This is in which part of the radio spectrum?
24. A scanning tunnelling microscope (STM) uses electron tunnelling across a vacuum gap to image surfaces at atomic resolution. Explain why the tunnelling current decreases exponentially as the tip-to-surface distance increases. Why does this give the STM atomic resolution in the vertical direction?

Chapter 49 Checkpoint

Before moving on to Chapter 50, confirm you can do each of the following.

I can...	See Section
Explain what “quantised” means and identify where quantisation appears in physics	49.1–49.2
Calculate photon energy using $E = hf = hc/\lambda$ and convert between eV and J	49.3
Explain the photoelectric effect and calculate maximum electron KE and threshold wavelength	49.4
Calculate de Broglie wavelengths for particles and explain the significance	49.5
Explain wave-particle duality and describe the double-slit electron experiment	49.6
Explain what a probability cloud (wave function) represents for atomic electrons	49.7
State the Heisenberg Uncertainty Principle and apply it to estimate minimum energies	49.8
Explain quantum numbers and the Pauli exclusion principle; connect to the periodic table	49.9
Describe stimulated emission and quantum tunnelling; explain why each has no classical equivalent	49.10
Compare quantum and classical physics and identify which applies in a given situation	49.11
Give examples of technology based on quantum mechanical principles	49.12
Identify and correct common misconceptions about quantum physics	49.13

Continue to Chapter 50: Nuclear Physics and Radioactivity →